

Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Motivation: Networks Meet the Exchange Rate

Two literatures that rarely talk to each other:

- ① **Production-network Phillips curves** (Rubbo 2024; La'O & Tahbaz-Salehi 2022): with input-output linkages, sector-level price stickiness generates a genuine inflation–output tradeoff for *almost every* price index — except one. A unique “divine coincidence” (DC) index, weighting sectors by Domar-scaled stickiness, restores the tradeoff-free policy prescription. All of this is derived in a **closed economy**.
- ② **Small open economy NK models** (Galí & Monacelli 2005 and successors): the exchange rate is a cost-push shifter and a policy instrument, but the production side is a single aggregate sector — no network structure, no heterogeneity in who is exposed to FX shocks.

This paper puts them together. Once imported inputs enter a multi-sector network, the exchange rate becomes a *second* cost-push channel that hits sectors asymmetrically depending on their import centrality and their position in the domestic supply chain. Does the DC index — or any analog — survive?

Why This Matters: The Policy Question

Motivating case: a small oil-exporting economy. Resource sector is upstream, import-intensive (machinery) and export-heavy; Manufacturing sits in the middle; Services is downstream, domestically oriented, and dominates final consumption.

The question a central bank actually faces:

- Float and target inflation $\Rightarrow$ exchange rate absorbs FX/commodity-price shocks, but at the cost of nominal volatility and possibly balance-sheet risk.
- Peg $\Rightarrow$ eliminates FX risk, but forces *all* adjustment onto domestic prices and output — amplified by whichever sectors are most import-exposed and most central in the network.
- Manage the float $\Rightarrow$ partially, but at what welfare cost relative to the theoretical optimum?

Rubbo's closed-economy answer (target the DC index) says nothing about *which* exchange-rate regime to combine it with, or how network position shapes the answer. That is the gap this paper fills.

- **La'O & Tahbaz-Salehi (2022, *Econometrica*)** — optimal monetary policy in production networks; efficient allocations require targeting a Domar-weighted index, not simple CPI. Closed economy.
- **Rubbo (2024, *Econometrica*)** — the paper we extend. Sectoral Calvo pricing + IO network $\Rightarrow$ unique DC index with *no* cost-push residual; welfare-optimal Taylor rule targets it.
- **Auer, Levchenko & Sauré (2019, *ReStud*)** — international inflation spillovers propagate through global IO linkages; empirical evidence that network position matters for cross-border pass-through.
- **Pasten, Schoenle & Weber (2020, *JPE*)** — heterogeneous price rigidity across sectors reshapes monetary transmission; motivates letting $\hat{\delta}_i$ differ by sector here too.
- **Fanelli & Straub (2021, *JPE*)** — theory of sterilized FX intervention as a second instrument alongside the policy rate.
- **Gopinath & Itskhoki (2022)** — dominant currency pricing; **Amiti, Itskhoki & Konings (2019, *QJE*)** — variable markups shape exchange-rate pass-through at the firm level.

Gap: no existing paper combines a multi-sector Calvo network *and* a genuine open-economy block (UIP, imported inputs, FX regimes) to ask whether the DC-targeting result survives.

Rubbo (2024): What Carries Over

Closed-economy system (N sectors, Calvo $\hat{\delta}_i$, IO network Ω):

$$\text{IS: } \tilde{y}_t = \mathbb{E}\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}\pi_{t+1}^C - r_t^{\text{nat}})$$

$$\text{MC: } \mathbf{mc}_t = \alpha \mathbf{w}_t + \Omega \mathbf{p}_t - \log \mathbf{A}_t$$

$$\text{NKPC: } \boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t$$

$$\text{Policy: } i_t = r_t^{\text{nat}} + \phi_\pi \pi_t^C + \phi_y \tilde{y}_t$$

Lemma 1: natural output = Domar-weighted TFP,
 $y_t^{\text{nat}} = \sum_i \lambda_i a_{it}$.

Proposition 1 (DC index): unique index with *no* cost-push:

$$\pi_t^{DC} = \rho \mathbb{E}\pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\Lambda^{DC}} \tilde{y}_t, \quad w_i^{DC} \propto \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Why DC is unique: $\mathcal{V} = \Delta(I - \Omega\Delta)^{-1}$ has rank $N-1$ on the space of cost-push shocks $\Rightarrow$ a one-dimensional null space $\Rightarrow$ exactly one index ϕ with $\phi^\top \mathcal{V} = \mathbf{0}^\top$.

One shock, one channel: in the closed economy, only TFP shocks $\boldsymbol{\chi}_t$ enter the cost-push term. That is exactly why a single index can null it out.

The open question we ask: with a *second* cost-push channel (the exchange rate), can the same trick still null out both simultaneously?

This Paper: Contribution and Approach

What we do:

- Extend Rubbo's network Phillips curve to a small open economy: imported intermediate inputs (with an import-cost-share matrix Ω^F), imported final consumption, UIP with a debt-elastic premium.
- Ask whether the DC index survives when the exchange rate is a second cost-push channel, and quantify the welfare cost of float / peg / managed-float regimes.
- Calibrate to a 3-sector oil-exporting economy (Resource $\rightarrow$ Manufacturing $\rightarrow$ Services).

Methodological choice: the model is built and solved as the *original nonlinear* system directly in Dynare — Cobb-Douglas cost minimization, the exact recursive Calvo pricing problem, CRRA/Frisch household FOCs, CES consumption aggregation — rather than substituting in a hand-linearized Phillips curve. Dynare computes the nonlinear steady state numerically and perturbs around it, so the log-linear coefficients (e.g. $\hat{\delta}_i$) come out of the solution rather than being assumed.

Model Setup: Households

Preferences (CRRA consumption, Frisch labor supply):

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

Consumption aggregation: CES nest between a domestic bundle and an imported bundle, home-bias weight ω , elasticity η ; the domestic bundle is itself Cobb-Douglas across the N sectors with shares β_i^H :

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad C_t^H = \prod_i \left(\frac{C_{it}}{\beta_i^H} \right)^{\beta_i^H}$$

Household FOCs (nonlinear, exactly as solved in Dynare):

$$\text{Euler: } C_t^{-\gamma} = \beta(1+i_t) \mathbb{E}_t \left[C_{t+1}^{-\gamma} / \Pi_{t+1}^C \right]$$

$$\text{Labor supply: } W_t / P_t^C = C_t^{\gamma} L_t^{\varphi}$$

Asset side: single internationally traded bond B_t^* ; no domestic bond needed (representative household, net-zero supply, redundant).

Model Setup: Firms and Calvo Pricing

Production (Cobb-Douglas, sector i):

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F}$$

Cost minimization $\Rightarrow$ nominal marginal cost is multiplicatively separable:

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Calvo pricing — fraction δ_i resets each period (*not* $\hat{\delta}_i$: that is derived, not primitive):

$$P_{it}^{\#} = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{\mathbb{E}_t \sum_s \rho^s (1 - \delta_i)^s Y_{it+s} \widetilde{MC}_{it+s}}{\mathbb{E}_t \sum_s \rho^s (1 - \delta_i)^s Y_{it+s} / P_{it+s}}$$

Exact nonlinear recursion (as coded in Dynare, real MC $\widetilde{MC} = MC/P$):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1 - \delta_i) \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon} X1_{i,t+1}$$

$$X2_{it} = Y_{it} + (1 - \delta_i) \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^{\#} / P_{it} = \frac{\varepsilon}{\varepsilon - 1} X1_{it} / X2_{it}$$

$$1 = (1 - \delta_i) \Pi_{it}^{\varepsilon-1} + \delta_i (P_{it}^{\#} / P_{it})^{1-\varepsilon}$$

Why this matters: log-linearizing this exact recursion is what *produces* Rubbo's adjusted parameter $\hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$ — it is a derived object, not a primitive of the model.

Model Setup: The Open-Economy Block

Exchange rate and imports (S_t : nominal ER, P_t^{F*} : foreign import price, law of one price):

$$P_t^{FH} = S_t P_t^{F*}$$

UIP with a debt-elastic premium (Schmitt-Grohé & Uribe 2003, for stationarity):

$$(1 + i_t) = (1 + i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t \left[S_{t+1} / S_t \right]$$

Net foreign assets (current-account identity):

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Trade flows:

$$EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}} \right)^{-\theta^*}, \quad IM_t = C_t^F + \sum_i \Omega_i^F \frac{MC_{it} Y_{it}}{P_t^{FH}}$$

Key mechanism: imported inputs enter every sector's marginal cost through Ω_i^F — the exchange rate now moves marginal cost *exactly like a productivity shock*, scaled by each sector's import exposure and propagated through Ω^H .

Policy Regimes and Shocks

Three FX regimes (only the policy block changes):

Regime	Rule	Exchange rate
Float	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual from UIP
Managed float	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Three shocks studied:

- Sectoral TFP shocks a_{it} , $AR(1)$, $\rho_a = 0.90$
- Import price shock p_t^{F*} , $AR(1)$, $\rho_{pF} = 0.85$ — the pure FX/commodity-price channel
- Foreign demand shock D_t^* , $AR(1)$, $\rho_D = 0.80$ — affects exports, hence NFA and the exchange rate

Solution Method: Nonlinear, Not Pre-Linearized

Why solve the nonlinear model directly, rather than hand-linearize first:

- The Calvo slope $\hat{\delta}_i$ and the DC weights are *outputs* of log-linearizing the true recursion — coding them as inputs risks baking in the answer.
- A genuine nonlinear model lets us go to order-2 perturbation, which is required for valid welfare comparisons across policy regimes (Kim-Kim / Woodford: first-order-accurate welfare rankings are not valid — variance terms enter the welfare function only at second order).

Steady state: solved numerically in Dynare (`initval + steady;`), not imposed. Two closed-form results still apply exactly and are used directly, not approximated:

- **Lemma 1 (exact):** $\tilde{y}_t = \sum_i \lambda_{D,i} [\log(Y_{it}/\bar{Y}_i) - \log A_{it}]$ — no shadow flex-price block needed.
- **Proposition 1 weights (theory-motivated):** DC weights built from $\lambda_{D,i}$ and $\hat{\delta}_i$, applied to realized inflation in the true nonlinear simulation.

Implementation: `open_economy_network.mod` (Dynare) $\rightarrow$ `run_all_regimes.m` (3 regimes, IRFs + moments)
 $\rightarrow$ `analysis.py` (figures below).

Calibration: A 3-Sector Oil-Exporting Economy

Parameter	Value	Parameter	Value
β (discount factor)	0.99	Ω_{21}^H (Manuf. $\leftarrow$ Resource)	0.20
γ (risk aversion)	1.00	Ω_{32}^H (Services $\leftarrow$ Manuf.)	0.25
φ (inv. Frisch elasticity)	2.00	$\Omega_1^F, \Omega_2^F, \Omega_3^F$ (import shares)	0.30, 0.10, 0.05
η (home/foreign elasticity)	1.50	$\beta_1^H, \beta_2^H, \beta_3^H$ (consumption shares)	0.05, 0.15, 0.80
$\beta^{F,tot}$ (import share, CPI)	0.10	$\delta_1, \delta_2, \delta_3$ (Calvo reset prob.)	0.75, 0.50, 0.25
ε (within-sector elasticity)	8.00	ψ (NFA debt-elastic premium)	0.020
θ^* (export price elasticity)	2.00	ϕ_π, ϕ_y, ϕ_s (policy)	1.50, 0.50, 0.30

Sector story: Resource is upstream, flexible-priced, import-intensive (machinery) and the main exporter. Manufacturing is the middle of the chain. Services is downstream, sticky-priced, and dominates final consumption — the classic setting for network amplification.

Network Properties: Centrality and DC Weights

Domestic supplier centrality (Domar weight):

$$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i$$

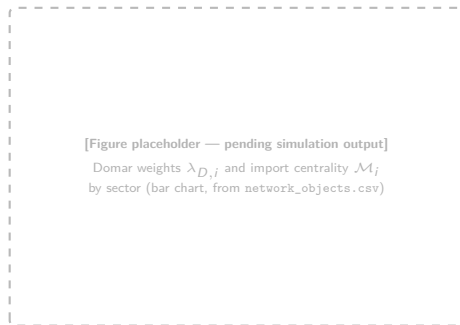
Import centrality (FX exposure transmitted through the domestic network):

$$\mathcal{M}_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i$$

DC-index weights (Proposition 1, applied here to the true nonlinear realized inflation):

$$w_i^{DC} \propto \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Both objects are computed once from the calibrated $\Omega^H, \Omega^F, \beta^H$ matrices — no magic numbers, same formulas as Rubbo's Proposition 1 and the import-centrality analog.



Expect: Services has the largest $\lambda_{D,i}$ (dominates final demand, downstream); Resource has the largest $\mathcal{M}_i$ (directly import-intensive) but this need not be the sector with the largest DC weight, since DC weights also load on stickiness $(1 - \hat{\delta}_i)/\hat{\delta}_i$.

Open-Economy Divine Coincidence: Does It Survive?

Closed economy (Rubbo): one cost-push channel (TFP) $\Rightarrow$ DC index exists uniquely because a rank-deficient matrix $\mathcal{V}$ has a one-dimensional null space, and one index can sit in it.

Open economy: two cost-push channels. TFP *and* the exchange rate now shift sectoral marginal cost. A single weight vector would need to null out *both* simultaneously:

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP channel}} \quad \text{and} \quad \underbrace{\phi^\top \Gamma = 0}_{\text{FX channel}}$$

Two conditions, one free vector $\Rightarrow$ generically no solution: **DC breaks down under a float.**

But: import nodes can be modeled as fully flexible upstream sectors ($\hat{\delta}_M = 1$) in an augmented network. Fully flexible nodes get *zero* DC weight ($w_M = 0$ since $(1 - \hat{\delta}_M)/\hat{\delta}_M = 0$), so Rubbo's Proposition 1 still applies to the *augmented* network — the DC index (as coded here) remains well-defined and insulates against domestic TFP shocks by construction. Whether it *also* nulls the FX channel in the calibrated network is the quantitative question tackled below.

Regime Comparison: Theoretical Predictions

	Float	Peg	Managed
Exchange rate	fully endogenous	fixed	partially stabilized
FX cost-push absorbed by	S_t adjustment	domestic π_i, y	split
Monetary independence	full	none (residual i_t)	full
Pass-through of import shocks	low (if ϕ_π high)	proportional to $\mathcal{M}_i$	intermediate
DC index well defined?	yes (augmented network)	n/a (S_t not a choice)	yes

Expected ranking (to be verified quantitatively): float should dominate on inflation stabilization when ϕ_π is high, at the cost of exchange-rate volatility; peg transmits import-price shocks directly, scaled by each sector's import centrality $\mathcal{M}_i$; managed float trades off between the two, with the trade-off governed by ϕ_s .

From Theory to Quantification

Pipeline: `open_economy_network.mod` solved under each regime → IRFs and order-2 moments exported → figures below.

Six questions the simulations answer:

- ① How do sectoral inflation, the output gap, and the exchange rate respond to TFP, import-price, and foreign-demand shocks, under each regime?
- ② Which sectors are most exposed, and does that line up with Domar weight, import centrality, or both?
- ③ How do exports and imports move, and what does that imply for the trade balance and NFA dynamics?
- ④ Does the DC index stay close to zero across shocks and regimes (insulation property), or does it visibly move under the float?
- ⑤ What is the welfare ranking of float vs. peg vs. managed float?
- ⑥ How much of the welfare cost is output-gap volatility vs. sector-level price dispersion?

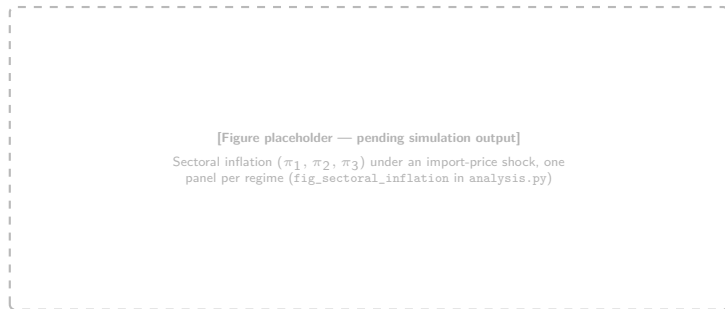
Results: Impulse Responses Across Regimes

[Figure placeholder — pending simulation output]

IRF grid: rows = shocks (TFP₁–TFP₃, import price, foreign demand); columns = CPI inflation, output gap, exchange rate, DC-index inflation; lines = float / peg / managed (`fig_irfs` in `analysis.py`)

What to look for: under the float, the exchange rate should absorb most of the import-price shock, leaving CPI inflation and the output gap comparatively flat; under the peg, the same shock should show up directly in sectoral and CPI inflation.

Results: Sectoral Inflation Heterogeneity



What to look for: under the peg, sectors should be ranked by import centrality $\mathcal{M}_i$ – Resource (most import-intensive) should show the largest and fastest inflation response; Services (least import-intensive, stickiest) should respond least but persist longest.

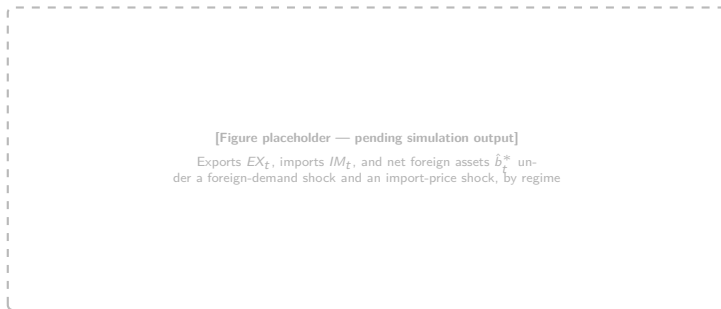
Results: Network and Import Centrality

[Figure placeholder — pending simulation output]

Domestic I-O matrix Ω^H heatmap, import-share bar chart, and import centrality $\mathcal{M}_i$ by sector (`fig_network` in `analysis.py`)

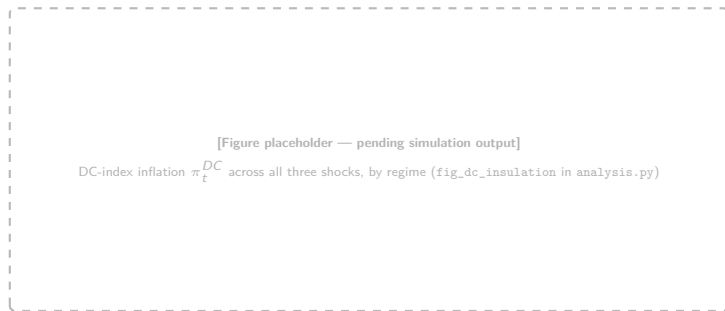
What to look for: the sector with the highest $\mathcal{M}_i$ (likely Resource, given direct 30% imported machinery share) versus the sector with the highest Domar weight $\lambda_{D,i}$ (likely Services, given its 80% final-consumption share) — these need not coincide, and the gap between them is exactly what makes network position matter for FX policy.

Results: Exports, Imports and the Trade Balance



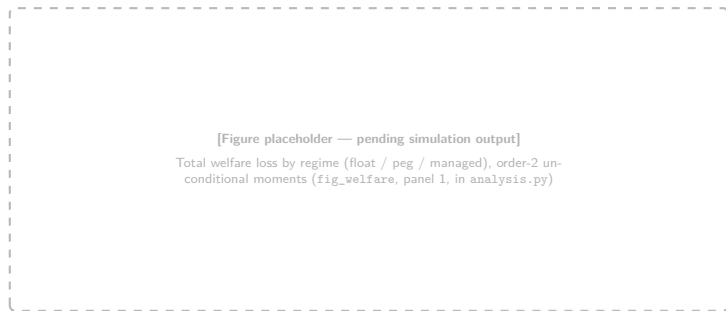
What to look for: a foreign demand shock should raise exports and NFA under all regimes, but by less under the peg if the associated appreciation pressure cannot be absorbed by S_t ; an import price shock should worsen the trade balance more under the peg, since the exchange rate cannot adjust to choke off import demand.

Results: Does the DC Index Insulate?



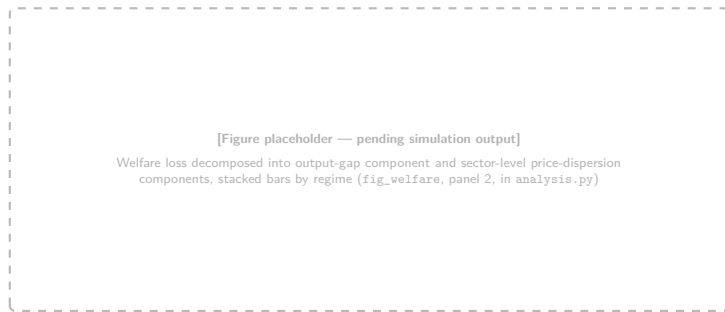
What to look for: under the float, π_t^{DC} should stay close to zero even for the import-price shock, since the exchange rate is left free to adjust and the Taylor rule directly targets π_t^{DC} ; any residual movement quantifies exactly how much the open-economy DC index fails to fully insulate (the FX-channel breakdown discussed above).

Results: Welfare Comparison Across Regimes



Welfare measure (Rubbo Proposition 3, adapted): $\mathcal{W} = \frac{1}{2} \left[(\gamma + \varphi) \text{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i} \varepsilon_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \text{Var}(\pi_{it}) \right]$, computed on the true nonlinear model's order-2 moments (not a linear approximation — first-order welfare comparisons across policy rules are not valid).

Results: Welfare Decomposition



What to look for: under the peg, expect the price-dispersion component to be concentrated in the most import-exposed, most centrally located sector; under the float, expect the output-gap component to be relatively larger (since inflation is well stabilized but exchange-rate adjustment itself has a real cost via the debt-elastic premium and terms-of-trade movements).

Summary: Key Messages

- ① Rubbo's divine-coincidence result relies on having exactly *one* cost-push channel. Opening the economy adds a second (the exchange rate), so a single index generically cannot null out both — **the DC-targeting prescription is not automatically inherited.**
- ② Treating imported inputs as fully flexible upstream nodes in an augmented network preserves a well-defined DC index that still insulates against domestic TFP shocks; whether it also insulates against FX shocks is a quantitative, not purely theoretical, question.
- ③ Network position matters for FX policy in two *distinct* dimensions: Domar centrality (who matters for aggregate output) and import centrality (who is exposed to the shock) — they need not coincide.
- ④ The welfare ranking of float / peg / managed float, and its decomposition into output-gap vs. price-dispersion costs, is calibration-dependent and is quantified directly from the nonlinear model's own second moments (order-2 perturbation), not a linearized proxy.

Remaining Work

- ✓ Full nonlinear structural model coded directly in Dynare
 - ✓ Numerical steady state; order-1 and order-2 perturbation solution
 - Done:** ✓ Network objects (Domar weights, import centrality, DC weights) derived, not hardcoded
 - ✓ Regime-switching infrastructure (float / peg / managed)
 - ✓ Analysis pipeline (IRFs, network plots, DC insulation, welfare decomposition)
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- ☐ Run `run_all_regimes.m`; verify Dynare steady state convergence
 - ☐ Populate all placeholder figures with actual simulation output
 - Remaining (target: July 22):** ☐ Sensitivity of welfare ranking to ϕ_s (managed-float sweep)
 - ☐ Sensitivity to import centrality calibration (alternative Ω^F)
 - ☐ Write up Sections 4–6 of the paper (quantitative results, discussion)